

10.2 Distinction TASK

About this task

Step-1

This task is designed to assess the D level expectations.

Step-2

Your tutor will then review your submission and will give you feedback. If your submission is incomplete the tutor will ask you to include missing parts. Tutor can also ask follow-up questions, either to clarify something that you have submitted or to assess your understanding of certain topics.

Feedback and submission deadlines

Feedback deadline: Friday 22 May (No submission before this date means no feedback!)

Submission deadline: Before creating and submitting portfolio.

Required documents

Execute your code into a jupyter notebook (.ipynb file) and keep the output, write a report (.pdf file) to explain what you do in each of the following steps, and submit your code and report to OnTrack.

The goal for this D task is to experiment with two semi-supervised learning algorithms and compare their performance with further error analysis.

Execute your code into a jupyter notebook (.ipynb file) and keep the output, write a report (.pdf file) to answer the following questions, and submit your code and report.

1. Load the MINIST data set, randomly sample 20% as the labelled set, 70% as unlabelled set and the rest 10% as the test set.
2. Train a feed forward neural net on the above 20% labelled training set, run self-training with the 70% unlabelled set. Can you get a converged model? If yes report the number of iterations you run with self training, if not improve the percentage of the labelled set and repeat the self training process until you can get a converged model. Report the final model performance with key findings.
3. Given the same setup, apart from self training, run another semi-supervised learning algorithm, compare the results and explain your key findings.
4. For the above two semi-supervised learning algorithms, do both quantitative and human error analysis on the wrong pseudo labels.